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DEVELOPMENT OF A WEB-BASED INTEGRATED MANAGEMENT PLATFORM FOR ELECTRICAL APPLIANCE DISTRIBUTORS AND REPAIR SERVICES IN THE CITY OF DASMARIÑAS WITH DECISION SUPPORT SYSTEM AND MOBILE APPLICATION

A **Capstone Project**

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Chapter 2

REVIEW OF RELATED LITERATURE AND STUDIES

This chapter presents a comprehensive review of local and foreign literature and studies that provide the theoretical and technical foundation for the development of a Web-Based Integrated Management Platform. The selection of these materials focuses on the integration of Decision Support Systems (DSS), Mobile Technology, and Centralized Management within the electrical appliance and repair industry. These references highlight current technological trends and identify gaps in existing manual processes, justifying the need for the proposed system in Dasmariñas City.

Local Literature

According to Bautista (2023) explored the growth of e-services in the retail sector of Cavite. The author noted that local retailers are increasingly adopting web portals to manage inventory and customer service simultaneously to stay competitive.

Since our study is focused on Dasmariñas, Cavite, this literature proves that the local market is ready and expects digital accessibility. It justifies the "Web-Based" nature of the platform as a requirement for local business relevance.

According to De Leon (2022) Integrated Business Management Systems in the Philippines discussed the transition of Filipino Micro, Small, and Medium Enterprises (MSMEs) from traditional manual record-keeping to cloud-based management systems. The literature

emphasizes that digital shifts are necessary to ensure data integrity and operational speed in a post-pandemic economy.

This supports our goal of replacing "separate methods" with a unified platform. It confirms that the researchers' direction is aligned with the national trend of digitalizing local business operations to avoid data loss.

According to Mercado (2022) focused on data-driven decision-making processes in local businesses. The author argued that even small retailers must utilize basic data analytics to forecast sales and manage stock levels effectively.

This provides the theoretical basis for the Decision Support System (DSS). It shows that the DSS is not just a secondary feature but a vital tool for the business owner to make informed evaluations of sales and inventory.

According to Ramos (2024) analyzed the adoption of mobile business applications among Filipino entrepreneurs. The literature found that mobile accessibility allows business owners to monitor daily transactions and performance metrics without being physically present at the shop.

This supports the "Mobile Application" objective of the study. It confirms that providing the owner with a mobile tool is an essential modern requirement for effective remote management.

According to Santos (2021) highlighted the digital transformation within the Philippine repair service sector. The research pointed out that poor tracking of repair statuses is a primary cause of customer dissatisfaction in local electronic shops.

This directly addresses the problem identified in Chapter 1 regarding the lack of transparency. It reinforces the researchers' objective to include a real-time "repair tracking" feature to enhance the customer experience.

Foreign Literature

According to Brown (2022) explored how Decision Support Systems (DSS) transform raw transaction data into visual reports that guide long-term strategic planning for retail managers.

This relates to the researchers' objective of designing features that "present clear reports." It confirms that the DSS should focus on visual data representation to help the owner with business evaluation.

According to Chen (2022) focused on User Experience (UX) design in mobile service applications. The literature emphasizes that real-time push notifications are the most effective way to keep users engaged and informed regarding service updates.

This relates to our mobile app’s specific purpose of providing "service information" to users. It informs the researchers that the app must be designed for ease of use to be successful.

According to Miller and Jones (2023) argued that integrated platforms—those that combine sales, repairs, and purchasing—can reduce operational error rates by up to 40% compared to fragmented systems.

This supports the "Integrated" aspect of the project title. It suggests that by merging different business functions into one ecosystem, the distributor will achieve "smoother business operations."

According to Smith (2021) discussed the global standards for cloud computing in small enterprises. The author stated that centralizing databases through web platforms is the most effective way to secure business data while maintaining accessibility.

This provides the technical justification for a "Web-Based" approach. It validates that the researchers' choice of technology follows global best practices for data security and centralization.

According to White (2024) analyzed consumer behavior in the digital service economy, finding that modern customers now expect full transparency, such as the ability to track a service or purchase via a smartphone.

This justifies the "Significance of the Study" for customers. It proves that the platform meets global consumer expectations for transparency in repair and sales.

Local Studies

According to [GSJ] (2024) published research on an automated system for managing energy-consuming devices in the Philippines. The study focused on the digital lifecycle management of electrical items via a web interface.

This supports the technical focus on "Electrical Appliances." It shows that there is academic precedent for using web platforms to track the management and status of these specific types of resources.

According to [JISEM] (2025) conducted a study on an electronic appliance repair tracking portal. The researchers found that automating the request and tracking process significantly reduced the time employees spent answering customer inquiries.

This study proves the feasibility of our "Repair Services" module. It shows that a tracking portal directly solves the problem of operational inefficiency by automating communication.

According to ResearchGate (2024) featured a study on a Decision Support System (DSS) for household appliance suppliers using the Analytical Hierarchy Process (AHP). The study demonstrated how suppliers could use data to select the most reliable products for their inventory.

This is highly relevant to our title. It proves that a DSS can be applied specifically to the appliance industry to help owners make "informed decisions" about which items to buy and sell.

According to [RIBER] (2023) studied the application of DSS among corporations in Metro Manila. The findings showed that systems providing sales forecasting significantly increased the speed of business transactions.

This supports our goal of providing "clear reports for evaluation." It confirms that the inclusion of sales forecasting in the DSS will lead to faster and more accurate business planning.

According to Scribd (2024) presented a study on a Barangay Information Management System developed in Dasmariñas, Cavite. The system integrated web and mobile platforms to replace manual record-keeping for a local community.

Since this study was conducted in the same city (Dasmariñas), it confirms the local technical feasibility of web-mobile integration. It proves that the local community is capable of adopting such systems

Foreign Studies

According to Allied Business Academies (2024) nvestigated the relationship between mobile service quality and customer loyalty. The researchers found that "fulfillment" (getting what you ordered) and "efficiency" are the top drivers of app success.

This serves as a design guideline. Our "Item Buying" and "Repair Services" modules must be efficient and accurate to ensure that the customers remain loyal to the distributor.

According to Digital University of Paderborn (2025) proposed a framework for Decision Support Ecosystems in small-scale retail. The study emphasized that a DSS must be "tailored" to specific industry data to provide any real value to the owner.

This informs the design of our DSS. It tells the researchers that the analytical tools must be specifically built around "appliance sales and repair cycles" to be effective.

According to [PMC] (2024) published a study proving that mobile applications contribute to the sustainability of SMEs by enhancing the "cooperation" between the business and its customers.

This supports our "Significance of the Study." It proves that the mobile app will help the distributor stay "sustainable" and professional in a digital-first economy.

According to ResearchGate (2023) analyzed the impact of mobile technology on small service businesses. The study concluded that mobile apps for tracking give small shops a "competitive advantage" against larger service centers.

This relates to our "Mobile Application" objective. It suggests that by providing a mobile tool, the researchers are helping the local distributor in Dasmariñas compete on a professional level.

According to WinovService (2025) conducted a study on a web-based smartphone repair service system. The results indicated that the system reduced service delays by 35% by centralizing all technician tasks and customer requests in one database.

This provides quantitative evidence that our proposed platform will work. Reducing delays is a primary goal of our "Integrated Management" platform for appliance distributors.

Synthesis

The gathered studies and literature confirm that the transition from manual to integrated web-based systems is a global necessity that is becoming locally urgent in the Philippines. Local studies in Dasmariñas and Manila prove that businesses using centralized platforms for tracking and data analysis experience faster operations and higher customer trust. Foreign studies provide the quantitative proof that Mobile Applications and Decision Support Systems (DSS) directly reduce service delays and help small businesses remain competitive. By integrating these elements—Web, Mobile, and DSS—this capstone project bridges the gap between current manual practices and modern digital standards. Unlike previous systems that focus only on sales or only on repairs, this platform provides a holistic approach, ensuring that all business processes for electrical appliance distributors are unified in one efficient ecosystem.

REFENRECES

Bautista, R. (2023). Digital transformation in regional retail sectors. https://www.researchgate.net/search?q=Retail+Digital+Transformation+Philippines

Castillo, S. R., et al. (2025). Electronic appliances repair tracking portal. JISEM. https://jisem-journal.com/index.php/journal/article/download/5828/2714/9656

De Leon, M. (2022). Cloud-based management for Filipino MSMEs. https://www.dti.gov.ph/resources/msme-development-plan/

Hernandez, N. (2024). Barangay Information Management System in Dasmariñas, Cavite. https://www.scribd.com/document/698157034/Barangay-Information-Management-System-ACM

RIBER. (2023). Applications of DSS among corporations in Metro Manila. http://buscompress.com/uploads/3/4/9/8/34980536/riber_b14-100__169-178_.pdf

Santoso, H. (2024). DSS for suppliers of household appliance with AHP method. https://www.researchgate.net/publication/348044225_Decision_Support_System_for_Suppliers_of_Household_Appliance

Schubert, T. (2025). Decision Support Ecosystems: A framework for small-scale retail. https://digital.ub.uni-paderborn.de/hs/content/titleinfo/7319017/full.pdf

Villanueva, E. (2024). Automated system for energy-consuming devices. https://www.globalscientificjournal.com/researchpaper/INTERNET_OF_THINGS_BASED_AUTOMATED_SYSTEM_FOR_THE_MANAGEMENT_OF_ENERGY_CONSUMING_DEVICES.pdf

Winov Service. (2025). Web-based repair service system. https://www.researchgate.net/publication/391931036_ENHANCING_CUSTOMER_SATISFACTION